

STATISTICS 652

LECTURE 2-26

REVIEW: FAMILY OF DISTRIBUTIONS $\mathcal{F} = \{F_\theta : \theta \in \Theta\}$ OR FAMILY OF PROBABILITY MEASURES $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$

k-dimensional or k-parameter exponential family

$$f(\underline{n}; \underline{\theta}) = \exp \left\{ \sum_{j=1}^k C_j(\underline{\theta}) T_j(\underline{n}) + d(\underline{\theta}) + S(\underline{n}) \right\} I_A(\underline{n})$$

COMMON EXPONENTIAL FAMILIES

- GEOMETRIC (θ)
- NEGATIVE BINOMIAL (r, θ)
- BERNOULLI (θ), BINOMIAL (N, θ)
- MULTINOMIAL ($N, \underline{\theta}$)
- POISSON (λ)
- EXPONENTIAL (λ), GAMMA (d, β)
- BETA (d, β)
- WEIBULL (d, β)
- NORMAL (μ, σ^2)
- MULTIVARIATE $N_p(\underline{\mu}, \Sigma)$

COMMON NON-EXPONENTIAL DISTRIBUTIONS

- DISCRETE UNIFORM
- HYPERGEOMETRIC
- CONTINUOUS UNIFORM
- PARETO
- DOUBLE-EXPONENTIAL
- LOCATION-SCALE CAUCHY OR T.

IMPORTANT STEP: BE ABLE TO RECOGNIZED WHEN A STATISTICAL PROBLEM FALLS WITHIN THE EXPONENTIAL FAMILY

⇒ UNIFYING THEORY WILL APPLY

EXAMPLES

(1) $X \sim \text{MULTINOMIAL}_m(N, \theta)$

IS A $k = (m-1)$ dimensional exponential family

DISCUSSED IN LAST LECTURE

(2) a) $X_1, \dots, X_n$ iid $\text{GAMMA}(\alpha, \beta)$ $\theta = (\alpha, \beta)$ $\alpha > 0, \beta > 0$

$$f(x; \alpha, \beta) = \left\{ \beta^\alpha \Gamma(\alpha) \right\}^{-n} \left(\prod_{i=1}^n x_i \right)^{\alpha-1} e^{-\sum x_i / \beta} \quad x_i > 0$$

is a $k=2$ dimensional exponential family (HW)

b) $X_1, \dots, X_n$ iid $\text{GAMMA}(\alpha(\theta), \beta(\theta))$ $\theta \in \mathbb{R}$ $\alpha: \mathbb{R} \rightarrow \mathbb{R}^+$ $\beta: \mathbb{R} \rightarrow \mathbb{R}^+$

$$f(x; \theta) = \left\{ \beta(\theta)^{\alpha(\theta)} \Gamma(\alpha(\theta)) \right\}^{-n} \left(\prod_{i=1}^n x_i \right)^{\alpha(\theta)-1} e^{-\sum x_i / \beta(\theta)}$$

$$= \exp \left\{ \alpha(\theta) \sum_{i=1}^n \log x_i - \frac{1}{\beta(\theta)} \sum_{i=1}^n x_i - n \log [\beta(\theta)^{\alpha(\theta)} \Gamma(\alpha(\theta))] \right\} = \sum_{i=1}^n \log x_i \Big\} T_A(x)$$

$$A = \{x \mid x_i > 0, i=1, \dots, n\}$$

FOR $\alpha(\theta) \neq \frac{1}{\beta(\theta)} \Rightarrow k=2$ dimensional exp. family

FOR $\alpha(\theta) = \frac{c}{\beta(\theta)} \Rightarrow k=1$ dimensional exp. family

DEFINITION 2.1. A k -dimensional exponential family is said to be of full rank

IF $\mathcal{C} = \{(\eta_1(\theta), \dots, \eta_k(\theta)) \mid \theta \in \Theta\}$ CONTAINS AN OPEN SUBSET OF $\mathbb{R}^k$.

OTHERWISE, IT IS CALLED A CURVED EXPONENTIAL FAMILY.

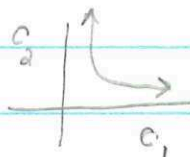
EXAMPLES: 1) MULTINOMIAL (N, θ) IS OF FULL RANK ($k=m-1$)

SINCE $\mathcal{C} = \{(\log \frac{\theta_1}{\theta_m}, \dots, \log \frac{\theta_{m-1}}{\theta_m}) \mid 0 < \theta_j < 1, \sum_{j=1}^m \theta_j = 1\}$ CONTAINS AN

OPEN SUBSET OF $\mathbb{R}^{m-1}$. NOTE: $\mathcal{C} = \mathbb{R}^{m-1}$

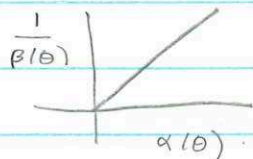
2) $X_1, \dots, X_n$ iid GAMMA (θ, θ) , $\theta > 0$ ($k=2$)

$$\mathcal{C} = \{(\theta, \frac{1}{\theta}) \mid \theta > 0\}$$



$\Rightarrow$ NOT FULL RANK, BUT CURVED

3) $X_1, \dots, X_n$ iid GAMMA $(\theta, 1/\theta)$, $k=1$, FULL RANK



$$\zeta(\theta) = \theta$$

4) $f(x) = c e^{-(x-\mu)^4}$ $\mu \in \mathbb{R}, x \in \mathbb{R}$

HW: SHOW THIS IS A CURVED k -dim. exp. family,
with $k = ?$.

RANDOM SAMPLE FROM EXPONENTIAL FAMILY

THEOREM 2.2,

If $X_1, \dots, X_n$ are iid $F \in \mathcal{F} = \{F_\theta : \theta \in \Theta\}$ with $\mathcal{F}$ being a k -dimensional exponential family defined with

$$C_j(\theta), T_j(\underline{x}), j=1, \dots, k, d(\theta), S(\underline{x}) \text{ and } A, \text{ then}$$

$\underline{X} = (X_1, \dots, X_n) \sim F_n \in \mathcal{F}_n = \{F_{n,\theta} : \theta \in \Theta\}$ with $\mathcal{F}_n$ being a k -dimensional exponential family defined with

$$C_j(\theta) \quad j=1, \dots, k, \quad d_n(\theta) = n d(\theta)$$

$$T_{n,j}(\underline{x}) = \sum_{i=1}^n T_j(x_i) \quad j=1, \dots, k, \quad S_n(\underline{x}) = \sum_{i=1}^n S(x_i) \text{ AND}$$

$$A_n = \{(\underline{x}_1, \dots, \underline{x}_n) \mid x_i \in A \quad i=1, \dots, n\} = A^n \times \dots \times A = A^n$$

NOTE: Given $\underline{X}$, $T_n(\underline{X}) = \{T_{n,1}(\underline{X}), \dots, T_{n,k}(\underline{X})\}$ IS SUFFICIENT FOR $\theta \in \Theta$.

WHY? FACTORIZATION THEOREM

THEOREM 2.3, FOR THE SETTING IN THEOREM 2.2,

$T_n(\underline{X})$ IS MINIMAL SUFFICIENT FOR $\theta \in \Theta$.

PROOF: USING THEOREM 6.2.13 IN CASELLA + BERGER,

$$\frac{f(\underline{x}; \theta)}{f(\underline{x}; \eta)} = \exp \left\{ \sum_{j=1}^k C_j(\theta) (T_{n,j}(\underline{x}) - T_{n,j}(\eta)) + (S(\underline{x}) - S(\eta)) \right\}$$

DOES NOT DEPEND ON $\theta \Leftrightarrow T_n(\underline{x}) = T_n(\eta)$ * OTHERWISE IT IS NOT OF DIMENSION k ,

$\Rightarrow T_n(\underline{x})$ IS MINIMAL SUFFICIENT

EXAMPLE: a) $X \sim N(\mu, \sigma^2)$ $\mu \in \mathbb{R}, \sigma^2 > 0$

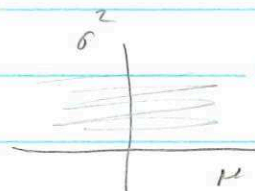
$\Rightarrow (X, X^2)$ IS MINIMAL SUFFICIENT NOTE: 1-1 FUNCTION OF X

b) $X_1, \dots, X_n$ iid $N(0, \sigma^2)$

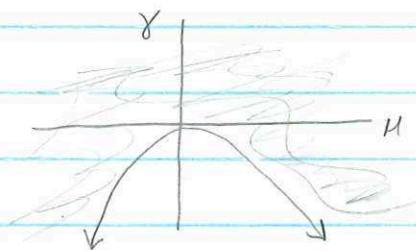
$\Rightarrow \sum_{i=1}^n X_i^2$ IS MINIMAL SUFFICIENT

NOTE: PARAMETRIZATION IS NOT UNIQUE.

i) $\text{NORMAL}(\mu, \sigma^2)$ $(H) = \{(\mu, \sigma^2) \mid \mu \in \mathbb{R}, \sigma^2 > 0\}$

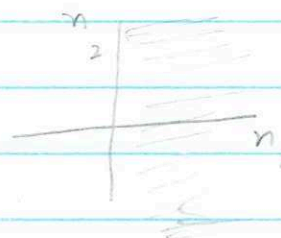


ii) $\text{NORMAL}(\mu, \mu^2 + \sigma)$ $(H) = \{(\mu, \sigma) \mid \mu \in \mathbb{R}, \sigma > -\mu^2\}$



NATURAL PARAMETERIZATION?

$$\eta = \{\eta_1 = 1/\sigma^2, \eta_2 = \mu/\sigma^2 \mid \eta_1 > 0, \eta_2 \in \mathbb{R}\}$$



GO TO NATURAL PARAMETER SPACE ON PAGE 6

NORMALIZING CONSTANT

$$f(\underline{x}; \theta) = \exp \left\{ \sum_{j=1}^k c_j(\theta) T_j(\underline{x}) + d(\theta) + S(\underline{x}) \right\} I_A(\underline{x})$$

IS A FACTED DENSITY OR MASS FUNCTION

$\Rightarrow$

$$\Rightarrow d(\theta) = \begin{cases} -\log \int_A \exp \left\{ \sum_{j=1}^k c_j(\theta) T_j(\underline{x}) + S(\underline{x}) \right\} d\underline{x} & \text{DENSITY} \\ -\log \sum_{\underline{x} \in A} \exp \left\{ \sum_{j=1}^k c_j(\theta) T_j(\underline{x}) + S(\underline{x}) \right\} & \text{MASS} \end{cases}$$

$$d_0(\eta) = d(\theta) = -\log \int_A \exp \left\{ \sum_{j=1}^k \eta_j T_j(\underline{x}) + S(\underline{x}) \right\} d\underline{x} \quad (\text{or } \sum_{\underline{x} \in A})$$

* NATURAL PARAMETER SPACE

$$\bar{\eta} = \{ \underline{\eta} = (\eta_1, \dots, \eta_k) \mid \underline{\eta} = (c_1(\theta), \dots, c_k(\theta)), \theta \in \Theta \}$$

$$f(\underline{x}; \underline{\eta}) = \exp \left\{ \sum_{j=1}^k \eta_j T_j(\underline{x}) + d_0(\underline{\eta}) + S(\underline{x}) \right\} I_A(\underline{x}) \quad \underline{\eta} \in \bar{\eta}$$

$$\text{where } d_0(\underline{\eta}) = d(\theta)$$

** GO TO NORMALIZING CONSTANT ON PAGE 5

NOTE: η IS A 1-1 FUNCTION OF θ . WHY? BY IDENTIFIABILITY

LEMMA 2.4. If $f(\underline{x}; \eta_1)$ and $f(\underline{x}; \eta_2)$ are valid p.d.f. (or p.m.f.)

and $\eta = \lambda \eta_1 + (1-\lambda) \eta_2$ with $0 \leq \lambda \leq 1$, then

$f(\underline{x}; \eta)$ is a valid p.d.f. (or p.m.f.).

THAT IS: $\bar{\eta}$ can be embedded into a convex parameter space.

EXAMPLE: $X \sim \text{GAMMA}(\theta, \theta)$

$$f(x; \theta) = \exp \left\{ (\theta-1) \log x - \frac{1}{\theta} x + d(\theta) \right\} I(x > 0) \quad \theta > 0$$

$$\text{where } d(\theta) = -\log \Gamma(\theta)$$

$$\eta_1 = \theta - 1, \eta_2 = -1/\theta$$

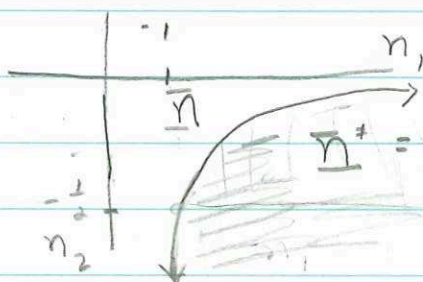
$$\text{or } f(x; \underline{\eta}) = \exp \{ \eta_1 \log x + \eta_2 x + d_0(\underline{\eta}) \} I(x > 0) \quad \underline{\eta} \in \bar{\eta}$$

$$\text{with } \bar{\eta} = \{ (\eta_1, \eta_2) \mid \eta_1 > -1; \eta_2 = -1/(1+\eta_1) \}$$

$$\text{and } d_0(\underline{\eta}) = -\log \left\{ \left(-\frac{1}{\eta_1} \right)^{\eta_1+1} \Gamma(\eta_1+1) \right\}$$

NOTE: ANY CURVED EXPONENTIAL FAMILY
CAN BE EMBEDDED IN A FULL RANK
EXPONENTIAL FAMILY.

CONVEX
COMPLETION



$$\bar{\eta}^+ = \{ (\eta_1, \eta_2) \mid \eta_1 > -1, \eta_2 \leq -1/(1+\eta_1) \}$$

NOTE: RESTRICTION ON
PARAMETER SPACE,

$$x = \eta_1 + 1 > 0, \quad \times \quad \times \quad \times$$